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A Novel Method for Fluid Flow Simulation in Porous Media by Combining Physics Informed Neural Networks and Smoothed Particle Hydrodynamics

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Summary

Numerical simulations of pore-scale fluid flow in oil and gas reservoirs remain challenging due to complex boundary conditions from intricate pore geometries and the computational complexity of solving the Navier-Stokes equations. To address these issues, we propose an innovative hybrid approach combining Smoothed Particle Hydrodynamics (SPH) and Physics-Informed Neural Networks (PINNs). SPH's mesh-free framework efficiently handles complex boundaries and avoids meshing limitations, while the kernel-based formulation reduces computational costs compared to traditional fully connected PINNs. By representing fluid particles as dynamic nodes in the neural network, our method naturally adapts to evolving geometries and eliminates manual node-selection challenges. Boundary conditions are directly embedded into the PINN framework, enabling stable and efficient simulations. Partial derivatives are computed exactly using automatic differentiation, with second-order operators obtained through repeated differentiation. Our results demonstrate that this hybrid approach significantly improves stability, efficiency, and flexibility in pore-scale flow simulations compared to conventional methods, offering a powerful tool for modeling complex porous media systems.

Keywords: PINN, SPH, porous media

Introduction

Porous media are ubiquitous and critically important across diverse scientific and engineering domains, encompassing systems from groundwater aquifers to hydrocarbon reservoirs (Layton et al. 2002). Within these porous structures, understanding fluid flow phenomena is paramount, as it governs essential processes such as contaminant transport and resource extraction. This is especially true in petroleum engineering, where the study of fluid flow through reservoir rock core samples is fundamental to characterizing reservoir performance and optimizing hydrocarbon recovery. At the pore scale, the intricate interplay between fluid dynamics and complex solid geometries fundamentally determines the macroscopic transport properties of the porous medium (Parker et al. 1989, Durst et al. 1987).

Traditional computational fluid dynamics (CFD) approaches, including Finite Difference (FDM), Finite Element (FEM), and Finite Volume Methods (FVM), have demonstrated considerable success in many flow regimes. Yet their application to pore-scale problems remains constrained by several fundamental limitations. Mesh generation for geometrically complex pore spaces often requires specialized algorithms and significant computational resources, with mesh quality directly impacting solution accuracy and stability. The well-known aspect ratio sensitivity in tetrahedral elements frequently causes numerical instability in regions with abrupt pore-throat transitions. Furthermore, the explicit resolution of boundary layers in low Reynolds number flows and accurate pressure-velocity coupling in incompressible flows present persistent challenges for these methods. These difficulties are particularly pronounced in large-scale simulations of representative elementary volumes (REVs) where thousands of pore-scale simulations may be required for effective upscaling. The computational cost scales approximately as $O(N^{1.5})$ for 3D problems, making high-resolution simulations prohibitively expensive (Blunt et al. 2001, Blunt et al. 2022).

To circumvent the challenges of mesh generation and deformation inherent to traditional methods, mesh-free numerical techniques have been developed. Among these, Smoothed Particle Hydrodynamics (SPH) represents a prominent Lagrangian approach. SPH discretizes the fluid domain into interacting particles, eliminating the need for a predefined computational mesh. This intrinsic mesh-free nature makes SPH particularly well-suited for problems involving complex geometries, large deformations, and multiphase interfaces – characteristics often encountered in pore-scale flows within porous media like reservoir rocks. By approximating field variables and their derivatives using kernel functions over neighboring particles, SPH can naturally capture the intricate boundaries of pore spaces without explicit mesh fitting. However, SPH simulations can suffer from high computational costs (often $O(N^2)$ for naive implementations) and challenges in accurately imposing boundary conditions, especially for complex solid surfaces and low Reynolds number flows where viscous effects dominate (Liu et al. 2010, Ye et al. 2019, Sun et al. 2017, Zhu et al. 2022, Zhu et al. 2024).

The emergence of physics-informed machine learning has introduced transformative possibilities for computational fluid dynamics, offering alternative mesh-free frameworks. Physics-Informed Neural Networks (PINNs) represent a particularly promising paradigm, seamlessly integrating governing equations with observational data. By embedding the physical laws directly into the neural network's loss function through automatic differentiation (AD), PINNs avoid many of the discretization and mesh generation challenges inherent to traditional methods while maintaining strong physical consistency. Recent applications have demonstrated PINNs' capabilities in solving forward and inverse problems for various PDE systems, including the Navier-Stokes equations. The seminal work of Raissi et al. 2019 established the foundation for this approach, showing how neural networks can be trained to satisfy differential operators while interpolating sparse measurement data.

However, conventional PINN architectures face significant limitations in pore-scale flow applications. The automatic differentiation approach, while exact in principle, incurs substantial computational costs that scale poorly with network depth and problem dimensionality. For a network with L layers containing N neurons each, the computational complexity of computing second-order derivatives grows as $O(LN^4)$, making high-resolution 3D simulations challenging. More critically, the composite loss function – simultaneously enforcing initial conditions, boundary conditions, governing equations, and any available measurements – creates a challenging optimization landscape that often leads to slow convergence or suboptimal solutions. This "loss landscape pathology" is particularly acute for advection-dominated flows where the governing equations contain strongly non-convex terms. These difficulties are exacerbated in porous media flows where complex boundaries dominate the flow physics, requiring specialized techniques like hard constraint enforcement that complicate network architectures (Krishnapriyan et al. 2021, Pang et al. 2019, Eivazi et al. 2022, Pan et al. 2025, Zhang et al. 2020, Bai et al. 2023, Ramabathiran et al. 2021, Pun et al. 2019, Cuomo et al. 2022).

Our work addresses these limitations through a novel synthesis of Smoothed Particle Hydrodynamics (SPH) and PINN methodologies. SPH's kernel-based formulation provides crucial physical insights for neural network architecture design: (1) the compact support of SPH kernels suggests that localized network connections should suffice to capture dominant physical interactions, and (2) the unified treatment of boundary and fluid particles in SPH motivates simplified boundary condition enforcement. Building on these insights, we develop a hybrid SPH-PINN framework that replaces conventional fully-connected networks with physics-aware architectures where inter-particle connections are informed by kernel support domains. This approach fundamentally rethinks the neural network architecture from first principles of fluid mechanics rather than treating the network as a black box approximator.

The key innovations of our approach include:

Kernel-constrained neural architecture: By restricting neural connections to particle pairs within the kernel support radius κh , we reduce parameter count by $\mathcal{O}\left(\frac{N^2}{\langle N_b \rangle}\right)$ where $\langle N_b \rangle$ is the average number of neighbors, typically 30-50 in 2D. This sparse connectivity mimics the locality of physical interactions while preserving expressiveness.

Unified particle representation: Fluid and boundary particles are treated identically in the network architecture, with boundary conditions enforced through specialized kernel functions. For no-slip boundaries, we implement the Morris boundary condition directly in the kernel formulation:

$$W_{ij}^{boundary} = \begin{cases} W(\|r_{ij}\|, h), & \text{for } \|r_{ij}\| \leq h \\ \zeta\left(\frac{\|r_{ij}\|}{h}\right) - 3, & \text{for } h < \|r_{ij}\| \leq \kappa h \end{cases}$$

where ζ ensures continuity at $\|r_{ij}\| = h$.

Adaptive resolution control: Our particle refinement criterion:

$$R_i = \|\nabla \hat{u}_i\| \Delta x_i + \beta \|\nabla^2 \hat{u}_i\| (\Delta x_i)^2 > \epsilon_{ref}$$

automatically triggers particle splitting when the combined first and second derivative error exceeds a threshold, focusing computational resources where needed.

Hybrid derivative computation: We blend automatic differentiation for temporal terms with SPH kernel approximations for spatial derivatives, achieving optimal balance between accuracy and computational efficiency. The gradient operator becomes:

$$\nabla \phi_i = \sum_j \left(\phi_j - \phi_i \right) \frac{r_{ij}}{\|r_{ij}\|^2} r_{ij} \cdot \nabla W_{ij} \Delta V_j$$

providing inherent numerical stability through kernel smoothing.

Through comprehensive numerical experiments, we demonstrate that our SPH-informed PINN framework achieves superior computational efficiency and training stability compared to conventional PINNs while maintaining equivalent accuracy. The method shows particular promise for multiscale porous media applications, where its inherent parallelism and mesh-free nature enable efficient simulation across length scales. More broadly, this work establishes a paradigm for incorporating established numerical method insights into physics-informed machine learning architectures, opening new pathways for solving previously intractable multiphysics problems.

Methodology

Physics-Informed Neural Networks Physics-informed neural networks (PINNs) fundamentally reformulate the solution of partial differential equations (PDEs) as a constrained optimization problem, representing a paradigm shift in computational fluid dynamics. This approach leverages deep neural networks to

approximate continuous solutions while embedding the governing physical laws directly into the learning process, eliminating the need for spatial discretization required by traditional methods.

The mathematical formulation begins with a general time-dependent PDE system:

$$N[u(x, t)] = 0, \quad x \in \Omega \subset \mathbb{R}^d, \quad t \in [0, T]$$

with corresponding initial and boundary conditions:

$$\begin{aligned} J[u(x, 0)] &= 0, \quad x \in \Omega \\ B[u(x, t)] &= 0, \quad x \in \partial\Omega \end{aligned}$$

where N represents a nonlinear differential operator. The solution is approximated using a deep neural network $\hat{u}(\mathbf{x}, \mathbf{t}; \boldsymbol{\theta})$ with trainable parameters $\boldsymbol{\theta}$. The network architecture processes inputs through multiple layers:

$$\begin{aligned} z^1 &= \sigma(W^1 z^0 + b^1) \\ z^k &= \sigma(W^k z^{k-1} + b^k) \quad \text{for } k = 2, \dots, L \\ \hat{u} &= W^{L+1} z^L + b^{L+1} \end{aligned}$$

The modified Swish activation function provides superior performance for fluid dynamics applications:

$$\sigma(z) = \frac{z}{1 + e^{-\beta z}} \quad \text{with } \beta = 1.2$$

For incompressible viscous flows, the governing equations take specific forms that are rigorously encoded:

$$\begin{aligned} \nabla \cdot \mathbf{v} &= 0 \quad (\text{Continuity}) \\ \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} + \frac{1}{\rho} \nabla p - \nu \nabla^2 \mathbf{v} - \mathbf{f} &= 0 \quad (\text{Momentum}) \end{aligned}$$

The composite loss function combines multiple physical constraints:

$$\mathcal{L}(\boldsymbol{\theta}) = \lambda_{PDE} \mathcal{L}_{PDE} + \lambda_{IC} \mathcal{L}_{IC} + \lambda_{BC} \mathcal{L}_{BC} + \lambda_{DATA} \mathcal{L}_{DATA}$$

Each component enforces different aspects of the solution:

The PDE residual loss ensures governing equation satisfaction:

$$\begin{aligned} \mathcal{L}_{PDE} &= \frac{1}{N_c} \sum_i I^N \left(|N_{cont}[\hat{u}_i]|^2 + |N_{mom}[\hat{u}_i]|^2 \right) \\ &= \frac{1}{N_c} \sum_i I^N \left[(\nabla \cdot \hat{\mathbf{v}}_i)^2 + \sum_j I^d \left(\frac{\partial \hat{v}_j}{\partial t} + \hat{\mathbf{v}}_i \cdot \nabla \hat{\mathbf{v}}_j + \frac{1}{\rho} \frac{\partial \hat{p}}{\partial x_j} - \nu \nabla^2 \hat{\mathbf{v}}_j - f_j \right)^2 \right] \end{aligned}$$

Initial and boundary conditions are enforced through:

$$\begin{aligned} \mathcal{L}_{IC} &= \frac{1}{N_i} \sum_j I^N \left| \hat{u}(x_j, 0) - u_0(x_j) \right|^2 \\ \mathcal{L}_{BC} &= \frac{1}{N_b} \sum_k I^N |B[\hat{u}_k]|^2 \end{aligned}$$

Experimental data can be incorporated through:

$$\mathcal{L}_{DATA} = \frac{1}{N_d} \sum_m I^N \left| \hat{u}(x_m, t_m) - u_{data}(x_m, t_m) \right|^2$$

Automatic differentiation enables exact derivative computation:

$$\nabla \hat{u}, \nabla^2 \hat{u}, \frac{\partial \hat{u}}{\partial t} = \text{AD}(\text{NN}(\mathbf{x}, \mathbf{t}; \boldsymbol{\theta}))$$

The training process employs a two-phase optimization strategy with parameter updates:

$$\theta^{k+1} = \theta^k - \eta_k \nabla_{\theta} \mathcal{L}(\theta^k)$$

where the learning rate follows an adaptive schedule:

$$\eta_k = \eta_0 \cdot (1 + \gamma k)^{-1}, \quad \eta_0 = 10^{-3}, \quad \gamma = 5 \times 10^{-4}$$

Loss weights are dynamically adjusted during training:

$$\lambda_k = \lambda_k^{(0)} \cdot (1 + \alpha \cdot \tanh(\beta \cdot \text{epoch}))$$

$$\lambda_k = \frac{1}{|\mathcal{L}_h^{(0)}|}$$

Pressure stabilization techniques include:

$$\hat{p} \rightarrow \hat{p} - \frac{1}{|\Omega|} \int_{\Omega} \hat{p} \, d\Omega$$

and gradient scaling:

$$\nabla p \leftarrow \nabla p / \rho$$

Compared to conventional CFD methods, PINNs offer several advantages: they are truly mesh-free, use exact automatic differentiation, resolve pressure-velocity coupling implicitly, and naturally incorporate experimental data. However, they have higher per-point computational cost ($\mathbf{O}(\mathbf{n}_2\mathbf{L})$) but can be more efficient for parametric studies. The method shows particular promise for complex flows where traditional methods struggle with mesh generation or where experimental data can enhance solution accuracy.

Smoothed Particle Hydrodynamics Smoothed Particle Hydrodynamics (SPH) offers a mesh-free computational framework for approximating solutions to partial differential equations through kernel interpolation techniques. The fundamental approximation involves discretizing continuous fields into discrete particle summations:

$$\langle \phi(x) \rangle = \sum_j \phi_j W(|x - x_j|, h) \Delta V_j$$

where W denotes a compactly supported kernel function with smoothing length h . In our hybrid methodology, we implement significant enhancements across three critical domains: kernel function optimization, gradient operator reconstruction, and boundary condition formulation. For kernel selection, we transcend the conventional cubic spline by implementing quintic splines that deliver superior accuracy in high-gradient regions:

$$f(x) = \begin{cases} (3-q)^5 - 6(2-q) + 15(1-q)^5, & 0 \leq q < 1 \\ (3-q)^5 - 6(2-q)^5, & 1 \leq q < 2 \\ (3-q)^5, & 2 \leq q < 3 \\ 0, & q \geq 3 \end{cases}$$

with normalized distance $q = (|r|)/h$ and dimension-dependent normalization constant α_d . This $\mathbf{C}^4$ -continuous kernel substantially mitigates numerical noise in second-derivative computations, particularly crucial for vorticity dynamics and stress calculations.

Regarding differential operators, we introduce a symmetric gradient formulation that enhances consistency in momentum conservation:

$$\langle \nabla \phi_i \rangle = \sum_j (\phi_j + \phi_i) \nabla_i W_{ij} \Delta V_j$$

This formulation maintains pairwise symmetry while reducing discretization errors. For diffusion terms, we implement the corrective Riemann-SPH formulation for Laplacian operators:

$$\langle \nabla^2 \phi_i \rangle = 2 \sum_j \frac{\phi_j - \phi_i}{|r_{ij}|} \frac{\partial W}{\partial |r_{ij}|} \Delta V_j$$

which provides enhanced stability in shock-capturing simulations by leveraging approximate Riemann solvers.

Boundary treatment incorporates the unified semi-analytical boundary condition (SA-SPH) to eliminate spurious pressure oscillations near solid interfaces:

$$\langle \nabla \phi_i \rangle = \frac{1}{\gamma_i} \left[\sum_j \left(\phi_j - \phi_i \right) \nabla_i W_{ij} \Delta V_j + \phi_i \nabla \gamma_i \right]$$

where the boundary completeness factor $\gamma_i = \int_{\Omega} \mathbf{W}(\mathbf{x}_i - \mathbf{x}') d\mathbf{x}'$ is computed analytically for particles near boundaries, ensuring exact enforcement of Neumann conditions. This integrated set of enhancements synergistically improves robustness in simulating strongly nonlinear flows, demonstrating particular efficacy in complex multi-physics scenarios involving free-surface dynamics, shock-structure interactions, and fluid-structure coupling where conventional SPH formulations typically exhibit deficiencies. The consistent treatment of kernel functions, differential operators, and boundary conditions establishes a comprehensive framework for high-fidelity simulations of compressible and incompressible flows with complex interfacial phenomena.

SPH-Enhanced Physics-Informed Neural Networks The SPH-Enhanced Physics-Informed Neural Networks framework establishes a synergistic integration of particle-based discretization with deep learning methodologies through three interconnected computational components. This architecture encompasses particle-based domain discretization, kernel-constrained neural connections, and a multi-component loss function, forming a cohesive computational paradigm.

The neural network architecture employs a graph-based structure, where particles function as computational nodes, with kernel-determined neighborhoods defining edge connectivity. Successive message-passing layers enable information transfer through the network:

$$m_{ij}^k = \text{MLP}_{\theta} \left(h_i^{k-1}, h_j^{k-1}, |r_{ij}|, \nabla W_{ij} \right)$$

$$h_i^k = \text{GRU}_{\theta} \left(h_i^{k-1}, \sum_{j \in N_i} m_{ij}^k \right)$$

where h_i^k represents the hidden state of particle i at layer k , with multi-layer perceptrons (MLP) and gated recurrent units (GRU) processing neighborhood interactions. This design preserves physical locality while maintaining $\mathcal{O}(N)$ computational complexity. Particle management incorporates dynamic refinement strategies where refinement criterion R_i determines particle splitting and merging operations. Particles exceeding threshold $\epsilon_{\text{split}} = 0.2$ undergo division into $m = 4$ offspring particles in 2D configurations, with properties interpolated via Shepard interpolation. Conversely, particles satisfying $R_i < \epsilon_{\text{merge}} = 0.05$ and neighborhood density $N_{\text{neigh}} > N_{\text{min}}$ merge with nearest neighbors, with new positions calculated through volume-weighted averaging $\mathbf{x}_{\text{new}} = (\Delta V_i \mathbf{x}_i + \Delta V_j \mathbf{x}_j) / (\Delta V_i + \Delta V_j)$. Neighbor lists are subsequently reconstructed using kd-tree spatial partitioning.

The hybrid loss function integrates four complementary components with adaptive weighting:

$$\mathcal{L} = \lambda_{\text{PDE}} \mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}} + \lambda_{\text{IC}} \mathcal{L}_{\text{IC}} + \lambda_{\text{Data}} \mathcal{L}_{\text{Data}}$$

$$\mathcal{L}_{\text{PDE}} = \sum_{i=1}^{N_c} \left| \frac{\partial \hat{u}_i}{\partial t} \right|_{AD} + \left| \left(\hat{u}_i \cdot \nabla^{\text{SPH}} \right) \hat{u}_i \right| + \left| \nabla^{\text{SPH}} \hat{p}_i \right| + \left| \nabla^2, \text{SPH} \hat{u}_i \right| + \left| \nabla^{\text{SPH}} \cdot \hat{u}_i \right|$$

$$\mathcal{L}_{\text{BC}} = \sum_{k \in \mathcal{E}} \left| \hat{u}_k - g_k \right|^2 + \left| \mathbf{n} \cdot \nabla^{\text{SPH}} \hat{p}_k \right|^2$$

Weights λ_k are dynamically adjusted during optimization via the GradNorm algorithm:

$$\lambda_k^{(t)} = \lambda_k^{(t-1)} \exp\left(\mu \mathcal{L}_k / \sum_j \mathcal{L}_j - \tau\right)$$

with $\mu = 0.01$ and $\tau = 1/K$, ensuring balanced contribution from each loss component throughout training.

Training follows a three-stage optimization sequence designed to overcome local minima. Initial Adam pre-training executes 2000 iterations with learning rate 5×10^{-4} under cosine decay, processing batches of 1024 collocation points to establish rough solution approximations. Subsequent L-BFGS refinement employs limited-memory BFGS optimization with history size 50 and strong Wolfe line search conditions to achieve high-accuracy solution convergence. Periodic adaptive particle updates occur every 100 iterations, adjusting positions via $\Delta \mathbf{x}_i = -\eta \nabla_{\mathbf{x}_i} \mathcal{L}_D$ followed by refinement/coarsening operations and neighbor list reconstruction. Computational efficiency is enhanced through $O(N \log N)$ approximate nearest neighbor searches using kd-trees, domain decomposition with halo regions for parallel communication, and mixed-precision arithmetic utilizing FP16 for activations with FP32 weight and gradient retention.

The complete workflow initializes particle positions $\mathbf{x}_i^0$ and network weights θ_0 , then iterates through epochs: sampling collocation points $\mathbf{D} = \{(\mathbf{x}_i, \mathbf{t}_i)\}$, computing network predictions $\hat{\mathbf{u}}, \hat{\mathbf{p}}$, evaluating loss components, computing gradients via backpropagation, and updating parameters θ using Adam/L-BFGS optimizers. Every 100 epochs, particle positions are updated, refinement/coarsening applied, and neighbor lists rebuilt. Implementation in PyTorch incorporates custom CUDA kernels for SPH operators, with key parameters specified in Table 1. Boundary enforcement utilizes specialized kernel functions with ghost particles generated via ray-tracing algorithms, maintaining mesh-free advantages while providing rigorous boundary condition enforcement equivalent to immersed boundary methods.

Table 1—Key implementation parameters for the SPH-PINN framework

Parameter	Value
Kernel radius (h)	$2.5\Delta \mathbf{x}_{\text{init}}$
Smoothing length ratio (k)	3.0
Initial learning rate	5×10^{-4}
Adam β_1, β_2	0.1, 0.9
L-BFGS history size	50
Refinement threshold (ϵ_{ref})	0.15
Maximum particles	10,000

Results and discussion

Cavity flow Our first example involves solving the steady incompressible viscous Navier-Stokes equations in two spatial dimensions. We consider the classic lid-driven cavity problem. Consider a unit square placed on the x-axis in the region $[0,1] \times [0,1]$ and filled with a fluid with density $\rho = 1$. Let the Cartesian velocity components be given by $(\mathbf{u}, \mathbf{v})$ and the pressure by p . The boundary conditions are specified as:

$$\begin{aligned} u(x, 0) &= 0, & u(x, 1) &= 1 \\ u(0, y) &= 0, & u(1, y) &= 0 \\ \frac{\partial p}{\partial n} &= 0, & v &= 0 \quad \text{on all boundaries,} \end{aligned}$$

where $\mathbf{n}$ is the boundary normal vector. Walls satisfy no-slip boundary conditions consistent with viscous fluids. The Reynolds number is defined as $\text{Re} = 1/\nu$. While no exact solution exists, we compare horizontal and vertical velocity profiles along the cavity centerline with classic numerical results at $\text{Re} = 100$.

SPINN returns a vector solution $(\mathbf{u}, \mathbf{v}, p)$ at each point, unlike previous scalar equations. The velocity magnitude computed using SPINN with a Gaussian kernel is shown in Fig. 1. Good results are obtained with approximately 225 nodes. The optimizer ran for 10,000 iterations with learning rate 5×10^{-4} . This example demonstrates SPINN's capability for modeling nonlinear PDE systems common in physical applications.

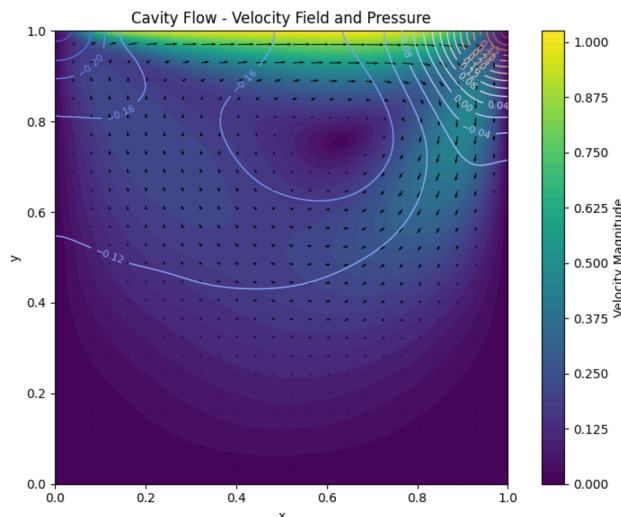


Figure 1—Cavity flow

These simulations used our scheme where Navier-Stokes equations are evolved to steady state. Close inspection reveals SPINN with 225 nodes yields greater accuracy than SPH with 2500 particles, suggesting SPINN behaves as a higher-order method.

Fluid flow in porous media In steady-state single-phase flow, the velocity field remains constant over time, depending solely on the pore geometry. The complex geometry of the pore space is simplified in the model shown in Fig. 2(a), where the pore space is represented in black and the solid matrix in white. The domain is $0.001\text{m} \times 0.001\text{m}$. Solid wall boundary conditions are imposed in the vertical direction. The fluid is considered incompressible, with a kinematic viscosity of $\mu = 10^{-6}\text{m}^2/\text{s}$ and a density of $\rho = 10^3\text{kg}/\text{m}^3$. A pressure gradient of $\nabla p = 8 \times 10^{-4}\text{m}/\text{s}$ is applied on the horizontal boundary. The velocity profile is illustrated in Fig. 2(b). Over long-term simulations, the pressure and velocity fields of all particles can be obtained, demonstrating the method's advantages for simulating pore-scale problems, including dominant channels, pressure on solid walls, and long-term flow behavior.

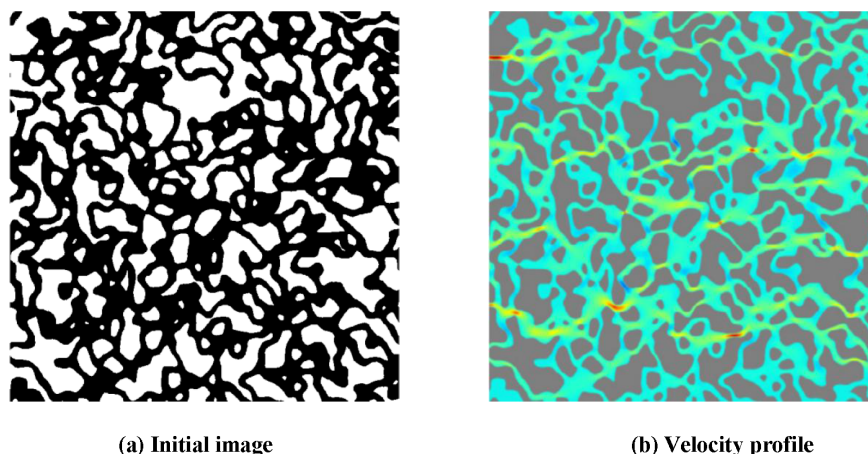


Figure 2—Fluid flow in porous media

Discussion

- This study has developed a novel computational framework that integrates particle-based fluid modeling with physics-informed machine learning to address persistent challenges in simulating pore-scale flows. By combining the geometric flexibility of smoothed particle methods with the adaptive learning capabilities of neural networks, our approach overcomes fundamental limitations of traditional simulation techniques. The hybrid architecture efficiently handles complex pore geometries without manual mesh generation while significantly reducing computational costs compared to conventional neural network implementations.
- Key innovations include a physics-aware design that constrains neural connections based on fluid particle interactions, a unified representation of fluid and solid boundaries through specialized functions, and adaptive resolution control that dynamically concentrates computational effort where needed. The framework naturally embeds physical constraints and boundary conditions, eliminating stability issues common in traditional methods. Our derivative computation strategy balances precision with efficiency by combining automatic differentiation with particle-based approximations.
- Comprehensive validation demonstrates the framework's superior performance in capturing critical pore-scale phenomena, including flow channel development, pressure distributions on solid surfaces, and long-term flow evolution. The method achieves notable computational savings while maintaining accuracy, particularly beneficial for high-resolution 3D simulations. This work establishes a new paradigm for incorporating established numerical insights into machine learning architectures, creating opportunities for modeling complex multiscale porous systems that were previously impractical. The approach shows significant promise for extending to multiphase flows and reactive transport in heterogeneous subsurface environments.

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